TEST PLAN

FarmFusion – Livestock Management Web App

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1. Introduction

FarmFusion is a web app (also usable on mobile screens) for managing livestock. After signing up and logging in, a user reaches a dashboard where they can buy and sell animals in a cattle market, track the progress of animals they bought, manage vaccinations, track breeding (including expected birth dates), and set up their cattle farm and animal types such as cows, bulls, sheep, goats and camels.

Application URL: https://farm-fusion-manage.vercel.app/login

2. Objective

To verify that the main features of FarmFusion (sign up, login, dashboard, cattle market, animal progress, vaccination, breeding and settings) work correctly, and to find and report defects before real users face them.

3. Scope

In scope:

- Sign up and login (valid, invalid and empty inputs)
- Dashboard and navigation between modules
- Cattle market: buying and selling animals
- Animal progress tracking for purchased animals
- Vaccination records and schedules
- Breeding records and expected birth dates
- Settings: cattle farm details and animal types (cows, bulls, sheep, goats, camels)
- Basic UI, mobile view and browser compatibility

Out of scope:

- Performance and load testing
- Full security penetration testing (only basic input checks)
- Real payment or delivery processing, if present

4. Features to Be Tested

Module	What will be tested
Sign Up	Registration with valid, invalid, empty and duplicate details; password rules
Login	Valid and invalid credentials, empty fields, password masking, logout and session
Dashboard	Page loads after login, summary information, links to every module
Cattle Market	Listing animals for sale, buying animals, required fields, price and quantity limits, invalid input
Animal Progress	Progress shown for bought animals, adding and updating progress details, correct animal data
Vaccination	Adding vaccination records, due dates and status, editing and deleting records, invalid dates

Module	What will be tested
Breeding	Breeding entries, expected birth date calculation, past and future date handling, editing records
Settings	Cattle farm details, adding and managing animal types (cows, bulls, sheep, goats, camels), saving changes
UI / Compatibility	Layout on mobile and desktop, browsers, spelling and alignment, error messages

5. Types of Testing

- **Functional testing:** each feature works as expected.
- **Negative testing:** wrong, empty or unexpected inputs.
- **Boundary testing:** minimum and maximum values, very long text, dates at the edges.
- **UI and compatibility testing:** mobile view, different browsers, layout.
- **Basic security testing:** special characters, access to pages after logout.

6. Test Environment

Item	Details
Application	https://farm-fusion-manage.vercel.app (deployed web app)
Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox
Devices	Desktop/laptop and mobile screen view
Test case tool	Microsoft Excel / Google Sheets
Defect tracking	Bug report files stored on GitHub
Version control	GitHub repository: farmfusion-qa-testing

7. Test Data

- Valid, invalid and already registered email addresses; strong and weak passwords
- Animal details for each type: cow, bull, sheep, goat and camel
- Valid, zero, negative and very large prices, ages and quantities
- Past, present and future dates for vaccination and breeding
- Very long text and special characters in text fields

8. Test Approach

Test cases are written module by module with an ID, steps, test data and expected result. Each test case is then executed manually on the live application, the actual result is recorded, and the status is marked Pass or Fail. Every failed test case is reported as a separate bug with steps to reproduce, expected and actual result, severity, priority and a screenshot.

9. Entry and Exit Criteria

Entry criteria:

- The application is deployed and reachable
- Test cases and test data are ready

Exit criteria:

- All planned test cases are executed

- Every failed test case is logged as a bug report

10. Test Deliverables

- Test Plan (this document)
- Test Cases (Excel/CSV)
- Bug Reports with screenshots
- README with project summary

11. Risks and Assumptions

- The application may be slow or unavailable because it is hosted online
- Test data may be limited for some animal types and dates
- Some expected results are based on the intended behavior of the app, and may be adjusted after discussing with the developer or team

12. Planned Test Coverage

The number of planned test cases per module is an estimate and may change as testing progresses. Execution results will be added after testing.

Module	Planned test cases
Sign Up	4
Login	5
Dashboard	3
Cattle Market (Buy / Sell)	8
Animal Progress	4
Vaccination	5
Breeding	5
Settings	6
UI / Compatibility	4
Total (approx.)	44